

# CS 330, Fall 2026, Homework 5

## Due Wednesday, October 7, 2026

*For full credit, you must follow the guidelines laid out on the first page of homework 1.*

### 1. Job Fair Speedrun

Boston University is hosting a massive Job Fair, and you are on a mission: hand out your CV and do an awkward-but-professional handshake with *every* employer. Unfortunately, every employer at the job fair has taken CS 330, so they are terrifyingly efficient with their time management.

Employer  $i \in \{1, \dots, n\}$  will only be at their booth during the time interval  $[s_i, f_i]$  (continuous time). If you show up at any time  $t \in [s_i, f_i]$ , you successfully shake hands with employer  $i$  (and possibly receive a sticker or a tote bag).

You want to find an *acceptable* set of visit times  $\{t_1, \dots, t_k\}$  such that each employer's interval  $[s_i, f_i]$  contains at least one  $t_j$  (i.e.,  $s_i \leq t_j \leq f_i$ ). Your goal is to minimize  $k$ , the number of visits to the job fair.

In the example of Figure 1, the black vertical lines mark a set of acceptable visit times. Removing any one of them would make the set not acceptable. (But it is still not the smallest acceptable set.)

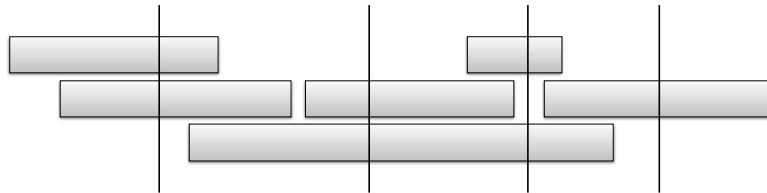


Figure 1: An acceptable set of four visit times.

- (Do not hand in) Find a smallest acceptable set of attendance times for the example in Figure 1.
- Your friend suggests a greedy strategy: always pick the time that covers the largest number of remaining employers. Give an example where this fails to find the minimum hitting set.
- Another friend observes: “If the input contains  $k$  mutually disjoint intervals, then every acceptable set of visit times must have size at least  $k$ .” Is she right? Prove or disprove.
- Design a polynomial-time algorithm that, given  $[s_1, f_1], \dots, [s_n, f_n]$ , outputs a minimum-size set of visit times. [*Hint*: modify the interval scheduling algorithm.]
- What is the running time of your algorithm in term of  $n$ ?
- Prove your algorithm is correct. [*Hint*: Use the lower-bound observation from part (c).]

### 2. Earliest Interval

Suppose you are given a set of closed intervals  $[s_i, f_i]$  for  $i = 1, \dots, n$ . For simplicity, assume that all start and finish times are distinct (that is, there are no repetitions in the list  $T = \{s_1, \dots, s_n, f_1, \dots, f_n\}$ ). For each finish time, we want to find the earliest-starting interval that

contains that finish time. More concretely, given  $f_k$ , find an index  $j$  such that

$$f_k \in [s_j, f_j],$$

and among all such intervals,  $s_j$  is as small as possible. Design an algorithm that outputs such an index for each finish time. The output may be given as  $A$ , a list of  $n$  indices such that for the  $k$ th finish time,  $A[k]$  is the value of the index you are asked to find.